Some commenters raised the more general concern that final rules could divert managers' attention from other types of risks that may be more urgent or important to investors.[3145] However, we expect this channel will be somewhat limited. First, the final rules will elicit more disclosures from those registrants for which climate-related risks have materially impacted or are reasonably likely to have material impacts on the registrants' financials or business strategy. Therefore, the final rules are unlikely to demand significant managerial attention in settings in which such attention is not warranted. Second, managers and directors have strong incentives to maximize the market value of the company (as reflected in the stock price). As a result, there is limited upside to selecting policies that prioritize climate over other concerns that investors view as more important determinants of company value.

2. Competition

Overall, we expect that by standardizing reporting practices, the final rules would level the playing field among firms, making it easier for investors to assess the climate-related risks of a registrant against those of its competitors. The effects of peer benchmarking can contribute to increased competition for companies in search for capital both across and within industries, whereby registrants can be more easily assessed and compared by investors against alternative options.

Some commenters raised concerns that the proposed rules would have increased competition among registrants for hiring individuals with climate-related expertise and/or GHG emissions attestation providers.[3146] These commenters asserted that the proposed rules could increase the costs of hiring key personnel with relevant experience, which could restrain a

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[3145] *See, e.g.*, letters from Chamber; Southside Bancshares; and BIO.

[3146] *See*, *e.g.*, discussions in sections II.E.2.b and II.I.5.b.

registrant's ability to produce climate disclosures and institute climate-related strategies.[3147] While the final rules do not completely eliminate concerns about the costs of hiring or engaging those with climate-related expertise, we have made several changes to mitigate these costs. With respect to GHG emissions assurance, for example, the final rules will permit assurance providers to use the ISO 14064-3 attestation standard, which should limit the circumstances in which registrants need to seek out different attestation engagements. In addition, the extended phase in periods for compliance with the GHG emissions disclosure and assurance requirements will provide additional time for registrants to seek out, and the markets to respond to increased demand for, climate-related professional services.

Some commenters stated that the proposed amendments would harm the competitive position of Commission registrants relative to their peers who do not face such disclosure requirements.[3148] In particular, these commenters stated that Commission registrants would face direct costs of compliance, and indirect costs such as the risk of disclosure of proprietary business information, while other companies would not face these costs.[3149] Relative to the proposed rules, the final rules take a number of steps to reduce the costs of complying with the final rules.[3150] For example, we have eliminated the requirement to disclose Scope 3 emissions, we have significantly narrowed the Regulation S-X requirements, and the final rules for subpart 1500 of Regulation S-K include additional materiality qualifiers and less prescriptive disclosure requirements. Moreover, as discussed above, a number of these changes from the proposal will

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3147 *See, e.g.*, letter from Can. Bankers

3148 *See* letters from API; Matthew Winden; and Southside Bancshares, Inc.

3149 *Id*.

3150 *See* section IV.C.2.

serve to limit the circumstances in which disclosure of potentially competitive business information will be required.

Similarly, one commenter noted that public companies could be placed at a competitive disadvantage when bidding to acquire a private target company because they would need to screen prospective targets for their ability to produce the disclosures required by the proposed rules.[3151] Any such competitive disadvantage will be mitigated under the final rules, as compared to the proposed rules, because we no longer are applying disclosure requirements to a private company that is a party to a business combination transaction, as defined by Securities Act Rule 165(f), involving securities offerings registered on Form S-4 or F-4.

Commenters also raised concerns about disproportionate effects for smaller companies, as discussed above in section IV.C.3.c. Any costs that disproportionately impact smaller companies—such as those that do not scale with the size of the registrant—may limit the ability of smaller registrants to compete with larger registrants. As discussed above, the final rules do not require SRCs and EGCs to provide GHG emissions disclosures and provide SRCs and EGCs with longer phase in periods to delay implementation costs. This delay may effectively lower implementation costs for SRCs and EGCs to the extent that, by the time they are required to report, SRCs and EGCs can look to the disclosure practices developed by other registrants to assist them in preparing their own disclosures.

[3151] Letter from Shearman Sterling. *See also supra* 2461 and accompanying text.

3. Capital Formation

More consistent, comparable, and reliable disclosures could lead to capital market benefits in the form of improved liquidity and lower costs of capital.[3152] These benefits would stem from reductions in information asymmetries brought about by the required disclosure of climate-related information.[3153] The reduction in information asymmetry between managers and investors could allow investors to better estimate future cash flows, which could reduce investors' uncertainty, thus lowering the costs of capital.[3154] In addition, less information asymmetry *among* investors could mitigate adverse selection problems by reducing the informational advantage of investors that have sufficient resources to become more informed about a registrant's exposure to and management of climate-related risks.[3155] This is likely to

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[3152] *See* D.W. Diamond & R.E. Verrecchia, *Disclosure, Liquidity, and the Cost of Capital*, 46 J. Fin. 1325 (1991) (finding that revealing public information to reduce information asymmetry can reduce a company's cost of capital through increased liquidity); *see also* C. Leuz & R.E. Verrecchia, *The Economic Consequences of Increased Disclosure*, 38 J. Acct. Res. 91 (2000). Several studies provide both theoretical and empirical evidence of the link between information asymmetry and cost of capital. *See, e.g.*, T.E. Copeland & D. Galai, *Information Effects on the Bid-Ask Spread*, 38 J. Fin. 1457 (1983) (proposing a theory of information effects on the bid-ask spread); Easley *et al.*, *supra* note 2753 (showing that differences in the composition of information between public and private information affect the cost of capital, with investors demanding a higher return to hold stocks with greater private information.).

[3153] *See, e.g.,* Christensen *et al.* (2021), at 1147 (noting "[A] primary benefit of corporate disclosure is to mitigate information asymmetries between the firm and its investors as well as among investors . . . [T]he general takeaway from this large literature is that more and better disclosure can lead to tangible capital-market benefits in the form of improved liquidity, lower cost of capital, higher asset prices (or firm value), and potentially better corporate decisions . . . To the extent that mandatory CSR reporting and CSR standards improve the information available to investors, the same theories and many of the prior findings should apply when considering the economic effects of the mandate or standard.").

[3154] *See* Diamond *et al.*, *supra* note 3152; Lambert, *et al.*, Accounting Information, *supra* note 2753; Christopher Armstrong, John Core, Daniel Taylor & Robert Verrecchia, *When Does Information Asymmetry Affect the Cost of Capital?*, 49 J. of Acct. Rsch. 1 (2011). We note that these articles also detail limited theoretical circumstances under which more reliable disclosures could lead to a higher cost of capital, such as in the case where improved disclosure is sufficient to reduce incentives for market making.

[3155] *See* Verrecchia, *et al.*, *supra* note 2748.

improve stock liquidity (*i.e.*, narrower bid-ask spreads), which could attract more investors and reduce the cost of capital overall.[3156]

There are two additional channels through which the disclosures could impact cost of capital. The first arises because some investors may have preferences to invest with companies that are more or less exposed to climate-related risks about which the final rules will elicit disclosure. To the extent the disclosures provide more complete and reliable information about a registrant's material climate-risks and how such risks are being managed, shifts in investor demand for the registrant's securities could increase or decrease (depending on investor preferences and how they factor this information into their investment decision-making).[3157] The

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[3156] One commenter asserted that this first channel does not apply to corporate disclosures, as it pertains only to bid-ask spreads set by market makers concerned with trading against parties with more information about order flow. *See* Overdahl exhibit to letter from Chamber. We disagree. Market makers concerned about trading against more informed parties will set larger bid-ask spreads regardless of the reason for the asymmetric information. In this setting, corporate disclosures of material climate-related information would reduce information asymmetries between market makers and other traders who have, for example, learned about a company's climate related risks through proprietary research. *See* letters from Calvert ("Calvert purchases third party vendor data to support our ability to assess companies on their ESG factors and that provide specific data related to climate change, where available. Often vendor information is estimated when a company has not disclosed information on its climate-related risks. Sometimes the estimates are made across industries, based on what other more proactive peers have disclosed."); Boston Trust Walden (reporting: "our analysts examine quantitative and qualitative climate-related corporate disclosure to enhance our understanding of the existing and potential financial outcomes associated, ranging from risks (e.g., losing the license to operate) to opportunities (e.g., generating new sources of revenue)"). We also note that corporate disclosures of material climate-related information reduce information asymmetries between affiliated investors and other investors. *See also* Glosten *et al.*, *supra* note 2748, for evidence that informed traders may take advantage of "private information or superior analysis" when making investment decisions). This commenter also asserted that the Commission must consider the potential efficiency losses that may result from investors no longer having the same incentives to invest in this type of proprietary research. We disagree with the commenter that there would be an efficiency loss. The primary benefit of proprietary research is more accurate prices. If disclosures obviate the need for proprietary research by achieving price discovery in the absence of that research, there is not an efficiency loss from the lack of research. This commenter also argues that voluntary disclosure regimes should enable corporate issuers to lower their cost of capital by reducing information asymmetry. *See supra* note 3154. We discuss shortcomings related to a voluntary disclosure regime in this context in section IV.B.2, and we cite to academic evidence in *supra* notes 2748 and 3153 that mandatory reporting that improves the information available to investors can lead to tangible capital market benefits.

[3157] *See* Yang, *supra* note 2827; Avramov, Cheng, Lioui & Tarelli, *Sustainable Investing with ESG Rating Uncertainty*, 145 J. of Fin. Econ. (Oct. 2022); L. Pastor, R. Stambaugh & L. Taylor, *Sustainable Investing in Equilibrium*, 142 J. Fin. Econ. 550 (2021); P. Bolton & M. Kacperczyk, *supra* note 2744; Li *et al.*, *supra* note 2657.

second results from the fact that some aspects of climate risk may not be diversifiable and therefore could command a risk premium. Academic research suggests that investors demand a higher return to hold assets that are more exposed to non-diversifiable climate-related risk (including both transition and physical risks).[3158] If the disclosures cause investors to update their expectations of a registrant's exposure to this type of risk, the cost of capital could adjust accordingly.

More generally, if compliance costs with the final rules are sufficiently high, this could influence the marginal company's decision to exit public markets or refrain from going public in the first place to avoid having to comply with the disclosure requirements. This concern was echoed by a number of commenters.[3159] Companies may choose this strategy if they believe the potential compliance costs from the final rules outweigh the benefits of being a registered public company including, for example, a more liquid market for the company's securities and the associated reduction in cost of capital. Uptake of this avoidance strategy may widen the transparency gap between public and private companies, negatively affecting capital markets' information efficiency, and potentially reducing the size of the public markets.[3160] However, we note that this avoidance strategy will come with significant disadvantages. For example, any companies deterred from registration because of the final rules would face more limited access to

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[3158] *See, e.g.,* Bolton *et al.*, *supra* note 3157 (finding that investors demand compensation for exposure to carbon emissions risk); Acharya *et al.*, *supra* note 2905 (finding higher expected returns for exposure to physical risks); Huynh & Xia (2021).

[3159] *See* letters from Elaine Henry; API; Cunningham *et al.*; Matthew Winden; Southside Bancshares Inc.; David Burton; AEPC; CCMR; Chamber; Petrol. OK; and AGs of Cal. *et al.*

[3160] *See* Overdahl exhibit to letter from Chamber.

the capital markets, implying higher financing costs and debt-ratios.[3161] On balance, we believe the benefits of being a public registered company are sufficiently strong such that it is unlikely many companies will choose to avoid becoming or continuing as a public registered company as a result of the final rules. In this regard, we note that the final rules include a number of changes from the proposal intended to mitigate the compliance burden on registrants and lessen disproportionate impacts on smaller and emerging growth firms.

F. **Reasonable Alternatives**

1. Adopt a more (or less) principles-based approach to Regulation S-K disclosures

Many commenters recommended a more principles-based approach (either overall or with respect to specific provisions) that would permit registrants to determine the type of climate-related information to disclose based on what they deem to be appropriate.[3162] Such an approach might reduce reporting costs because registrants would be required to report only information that they determine to be appropriate given their unique circumstances. To the extent that the more prescriptive elements of the final rules result in disclosure that is less useful for investors, a principles-based approach could benefit investors by reducing the incidence of less material or even boilerplate disclosure.[3163] A principles-based approach would also reduce the risk that the disclosure requirements could lead registrants to change their risk management

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[3161] *See* Omer Brav, *Access to Capital, Capital Structure, and the Funding of the Firm*, 64 J. of Fin. 263 (2009); Anthony Saunders & Sascha Steffen, *The Costs of Being Private: Evidence from the Loan Market*, 24 Rev. of Fin. Stud. 4091 (2011); E.P. Gilj & J.P Taillard, *Do Private Firms Invest Differently than Public Firms? Taking Cues from the Natural Gas Industry*, 71 J. of Fin. 1733 (2016).

[3162] *See*, *e.g.*, letters from Beller, *et al.* and Microsoft; Sullivan Cromwell; Airlines for America; BOA; Business Roundtable; Soc. Corp. Gov; and Overdahl exhibit to letter from Chamber.

[3163] Similarly, one commenter described "1) the ability of a principles-based approach to evolve in order to keep pace with emerging issues; and 2) the flexibility of a principles-based approach to correct deficiencies or excesses in disclosure without the need for the Commission to continuously add to or update the underlying disclosure rules as new issues arise." Overdahl exhibit to letter from Chamber. We acknowledge that a principles-based approach can present these benefits and that prescriptive rules may need updates.

strategies in ways that are less than optimal for the sake of achieving what they perceive to be more favorable climate-related disclosure.

On the other hand, a more principles-based approach would not fully achieve many of the intended benefits of the rules, which are focused on enhancing the consistency and comparability of existing voluntary disclosure arrangements. In addition, a principles-based approach could increase shareholder confusion because the choice of climate metrics and other details (e.g., time horizon) may vary significantly across registrants. Also, a principles-based approach may allow registrants to selectively choose the measures or time horizon that result in the most favorable disclosures. In the final rules, we elected to include prescriptive disclosure requirements (with certain modifications to address commenter concerns) to avoid such cherry-picking of information and to ensure that investors are provided with more consistent and comparable information about climate-related risks.

We similarly considered whether the final rules should be more prescriptive. This would generally improve investors' ability to compare disclosures across registrants since disclosures would be less tailored to each registrant's specific circumstances. A more prescriptive approach would also reduce the risk of boilerplate disclosures. However, we decided against this approach in light of commenters' concerns about the costs of compliance with the proposed rules, as well as the importance of allowing registrants the flexibility to provide investors with the most useful and relevant disclosures. Accordingly, in response to commenters, the final rules include additional materiality qualifiers and take a less prescriptive approach in a number of areas, which should help to mitigate some of the concerns expressed with respect to the proposed rules while continuing to elicit more decision-useful information for investors about climate-related risks.

2. Different approaches to assurance over GHG emissions disclosures

We considered several alternative approaches to assurance over GHG emissions disclosure. For example, the Commission could not require that any GHG emissions disclosure be subject to assurance. Alternatively, the Commission could require reasonable assurance of all GHG emissions disclosures rather than only for LAFs. The Commission could also prescribe more restrictive requirements for attestation standards and assurance providers. Inherent in these choices is a tradeoff between compliance costs and the reliability of the disclosures. For example, while requiring reasonable assurance for all GHG emissions would have likely resulted in more reliable disclosures, it would have imposed considerable costs on registrants, based on feedback from commenters about the costs of obtaining reasonable assurance.[3164]

We also considered taking a less prescriptive approach to the independence requirements for assurance providers in the final rules. For example, we considered not adopting a requirement for the GHG emissions assurance provider to be independent with respect to the registrant and any of its affiliates and/or instead requiring disclosure about any potentially independence-impairing relationship.[3165] This approach would help to mitigate concerns commenters raised about a potential shortage of qualified GHG emissions assurance providers increasing the costs for registrants[3166] and potential burdens on registrants related to the need to assess the independence of assurance providers.[3167] However, not imposing an independence requirement or only requiring disclosure about potential conflicts would not provide the same

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[3164] *See, e.g.*, letter from Salesforce (estimating that obtaining reasonable assurance rather than limited assurance over their emissions disclosures would increase their expected costs by $1-$3 million).

[3165] *See* 17 CFR 229.1506(b)(2).

[3166] *See, e.g.*, letters from AEPC; Climate Risk Consortia; and Soc. Corp. Gov.

[3167] *See, e.g.*, letter from Soc. Corp. Gov.

confidence to investors that the attestation provider will perform the engagement in an objective and impartial manner. This in turn would diminish one of the key benefits of requiring assurance over GHG emissions disclosures, which is to improve the reliability of such disclosures.

We acknowledge that the independence requirement in the final rules may result in some registrants that are already obtaining assurance voluntarily needing to retain a new GHG emissions assurance provider that meets the independence requirement or may make it more difficult for a registrant that has not obtained GHG assurance before to find an available provider. These costs are mitigated by the modifications in the final rules that provide registrants subject to the assurance requirement with a multi-year phase in period before they are required to obtain an attestation report. The phase in period will give registrants time to find a provider that meets the independence requirement or provide existing service providers time to unwind any existing conflicts in order to meet the independence requirement. It will also give non-accountant attestation providers time to familiarize themselves with the independence requirement and adapt their business practices accordingly.

3. Different thresholds for financial statement disclosures

We considered alternative criteria for disclosure under the amendments to Regulation S-X, such as using a more principles-based materiality approach. In general, materiality thresholds can help ensure that the disclosure elicited is most likely to factor into an investor's decision or voting decisions. While materiality is used as the threshold for disclosures in certain contexts, we believe that registrants will benefit from the certainty associated with a set of bright line quantitative thresholds. In doing so, investors will have disclosures that are more consistent across registrants due to the predictable application of quantitative thresholds. As discussed above, we have significantly modified the scope of the proposed disclosures and threshold and